Digital Integrated Circuit Design

Lecture-2: Review of Verilog HDL – Supplement (Memory)

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Outline

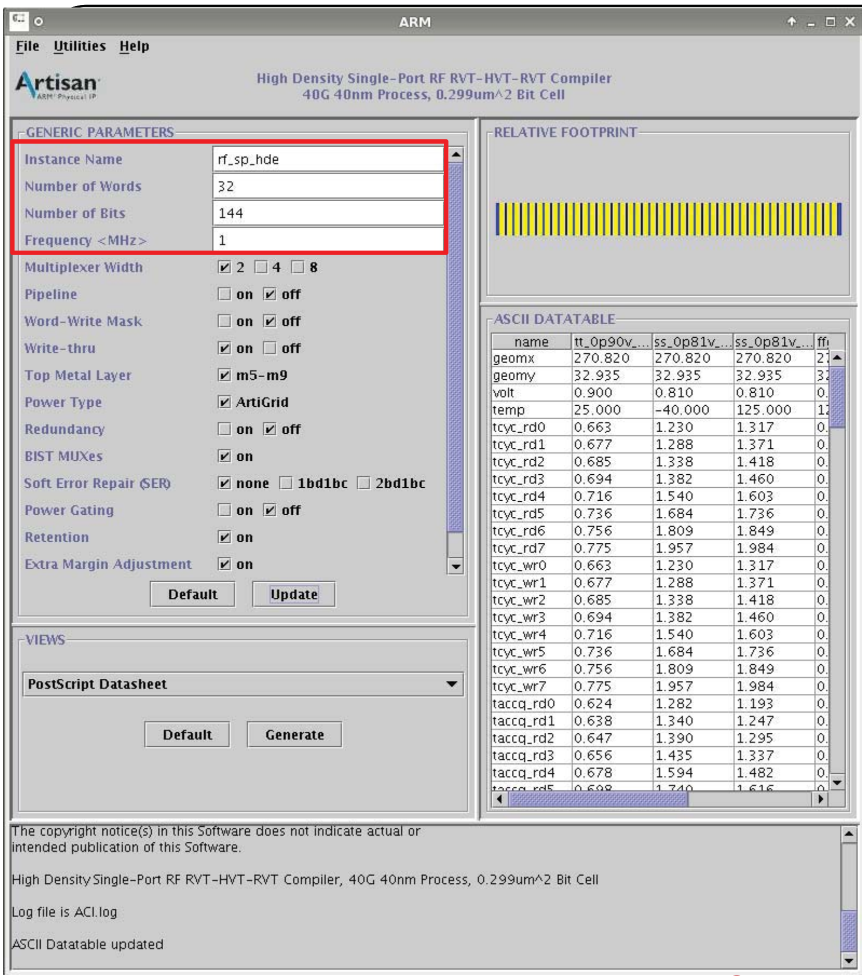
- SRAM and Register File (RF)
- Case Study – Memory Access Control with FSM

Register File (RF) and SRAM

- Register file
  - Built as a small, multiported SRAM array because it is more compact than an array of flip-flops
  - Suitable for small size of storage
- SRAM
  - Suitable for large size of storage

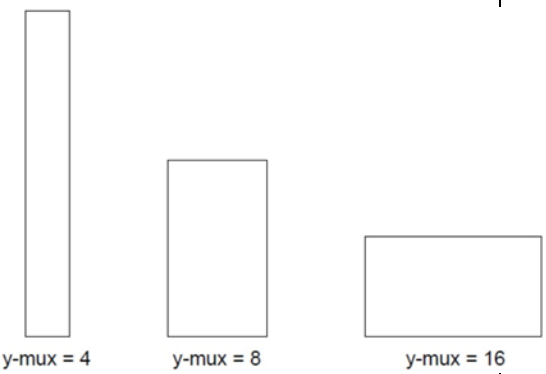
Register File (RF) and SRAM (cont'd)

- Port
  - Single-port
    - 1-WR or 1-RD in a clock
  - Two-port:
    - 1-RD, 1-WR, or 1-WR & 1-RD in a clock
    - 1 I/P bus, 1 O/P bus, and 2 address buses
  - Dual-port:
    - 2-RD, 2-WR, 1-WR & 1-RD in a clock
    - 2 I/P buses, 2 O/P buses, and 2 address buses
- Synchronous
  - Access SRAM/RF data with synchronous clock



Memory Complier

The selection of y-mux will affect the Memory characteristics such as access time, power consumption and area.



SRAM and Register File (cont'd)

- Word and Bit Selection

Parameters			YMUX = 2	YMUX = 4	YMUX = 8	YMUX = 16	YMUX = 32
Words (w)		Min	4	8	16	32	64
		Max	512	1024	2048	4096	8192
		Step	2	4	8	16	32
Bpw (b)	ba = 1	Min	1	1	1	1	1
		Max	128	64	32	16	8
		Step	1	1	1	1	1
	ba = 2	Min	2	2	2	2	2
		Max	256	128	64	32	16
		Step	1	1	1	1	1

- ba: Bank of memory

SRAM and Register File (cont'd)

- SRAM

	Multiplexer Width	Max word	Min word	Max bits	Min bits
1 port	8	4096	256	144	4
1 port	16	8192	512	72	4
1 port	32	16384	1024	36	4

- Register file

	Multiplexer Width	Max word	Min word	Max bits	Min bits
1 port	8	4096	256	144	4
1 port	16	8192	512	72	4
1 port	32	16384	1024	36	4

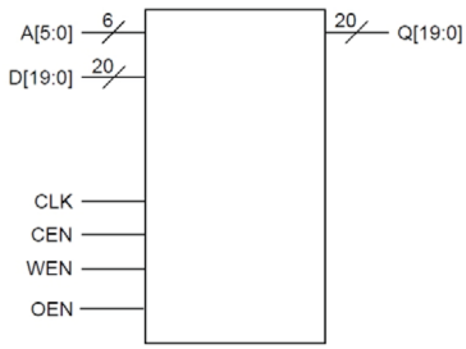
Memory

- High-speed single-port synchronous SRAM (0.18 μm CL018G CMOS process)
  - RA1_64×26 b (64×20, Mux 4, Drive 12)

Pin Description

Pin	Description
A[5:0]	Addresses (A[0] = LSB)
D[19:0]	Data Inputs (D[0] = LSB)
CLK	Clock Input
CEN	Chip Enable
WEN	Write Enable
OEN	Output Enable
Q[19:0]	Data Outputs (Q[0] = LSB)

Symbol



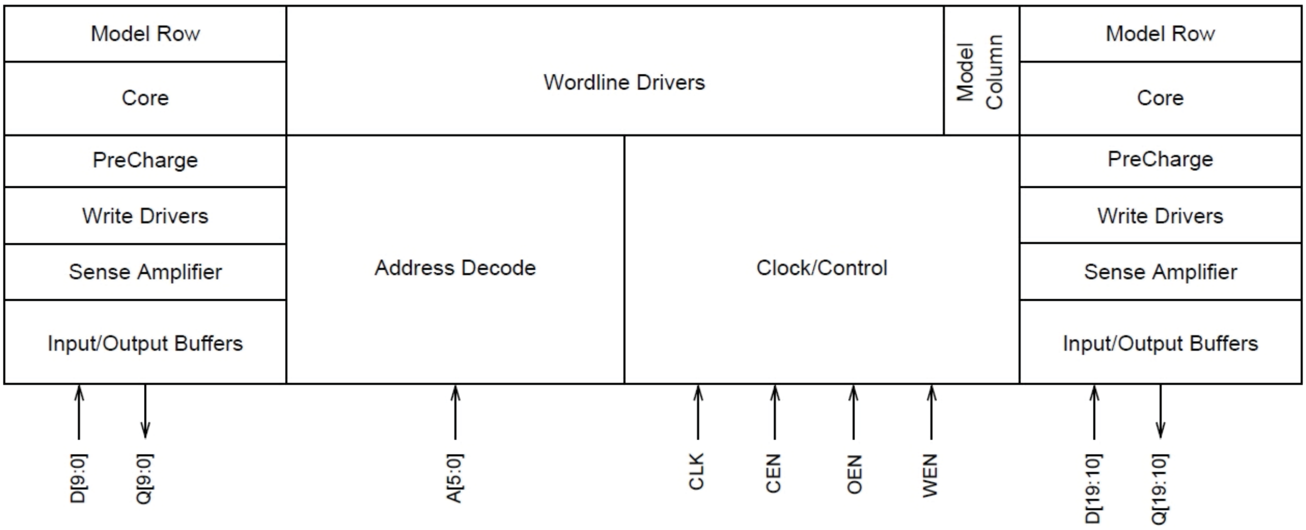
Area

Area Type	Width (mm)	Height (mm)	Area (mm^2)
Core	0.233	0.177	0.041
Footprint	0.243	0.188	0.046

The footprint area includes the core area and user-defined power ring and pin spacing areas.

Memory (cont'd)

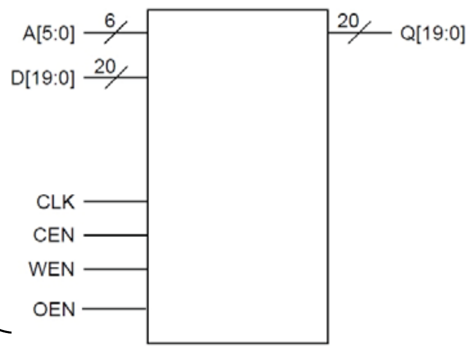
- Block diagram



Memory (cont'd)

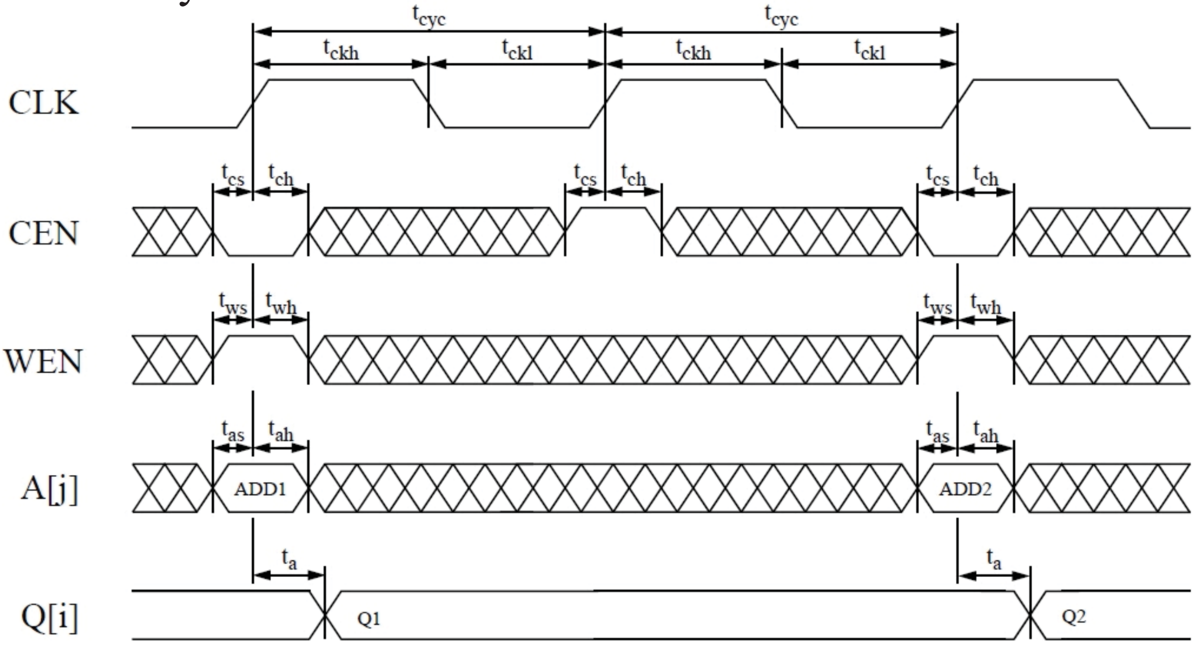
- SRAM logic table

CEN	WEN	OEN	Data Out	Mode	Function
H	X	L	Last Data	Standby	Address inputs are disabled; data stored in the memory is retained, but the memory cannot be accessed for new reads or writes. Data outputs remain stable.
L	L	L	Data In	Write	Data on the data input bus D[n-1:0] is written to the memory location specified on the address bus A[m-1:0], and driven through to the data output bus Q[n-1:0].
L	H	L	SRAM Data	Read	Data on the data output bus Q[n-1:0] is read from the memory location specified on the address bus A[m-1:0].
X	X	H	Z	High-Z	The data output bus Q[n-1:0] is placed in a high impedance state. Other memory operations are unaffected.



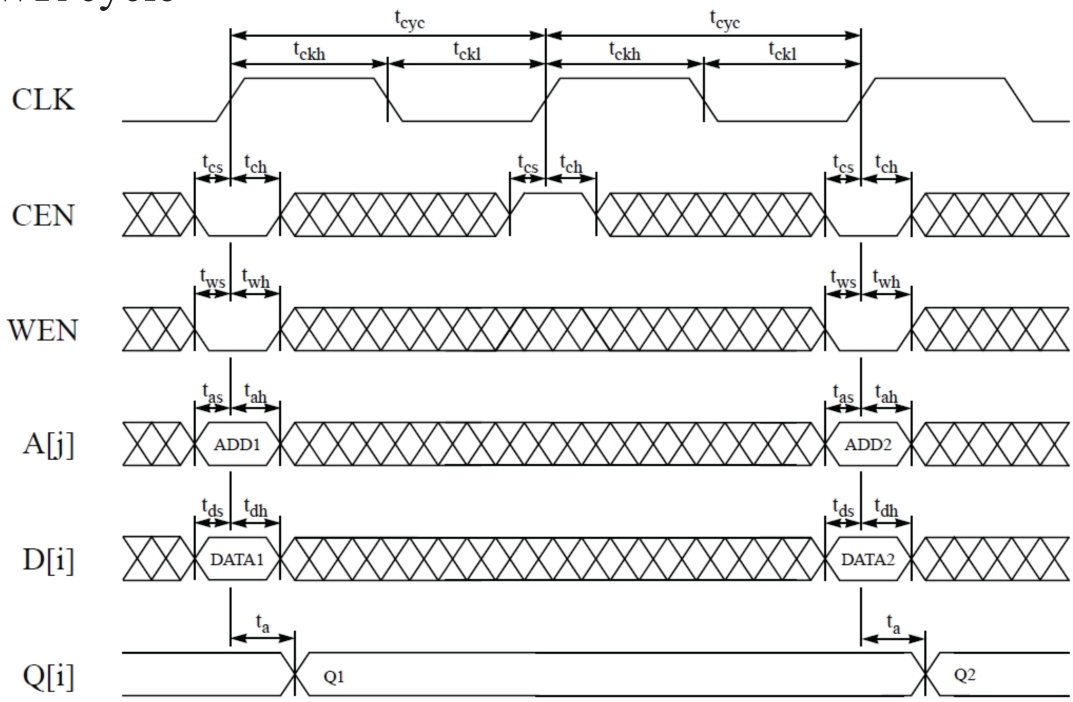
Memory (cont'd)

- Read cycle



Memory (cont'd)

- WR cycle



SRAM Timing

Parameter	Symbol	Fast@-40C Process 1.98V, -40°C		Fast@0C Process 1.98V, 0°C		Typical Process 1.80V, 25°C		Slow Process 1.62V, 125°C	
		Min (ns)	Max (ns)	Min (ns)	Max (ns)	Min (ns)	Max (ns)	Min (ns)	Max (ns)
Cycle time	t_{cyc}	0.74		0.79		1.14		2.02	
Access time[1,2]	t_a	0.73			0.78		1.22	2.10	
Address setup	t_{as}	0.17		0.18		0.27		0.51	
Address hold	t_{ah}	0.03		0.05		0.05		0.12	
Chip enable setup	t_{cs}	0.25		0.26		0.35		0.56	
Chip enable hold	t_{ch}	0.00		0.00		0.00		0.00	
Write enable setup	t_{ws}	0.24		0.24		0.33		0.55	
Write enable hold	t_{wh}	0.00		0.00		0.00		0.00	
Data setup	t_{ds}	0.11		0.11		0.17		0.32	
Data hold	t_{dh}	0.00		0.00		0.00		0.00	
Output enable to hi-Z	t_{hz}		0.42		0.43		0.59		0.93
Output enable active[1]	t_{lz}		0.37		0.38		0.52		0.82
Clock high	t_{ckh}	0.07		0.08		0.11		0.19	
Clock low	t_{ckl}	0.10		0.10		0.16		0.28	
Clock rise slew	t_{ckr}		4.00		4.00		4.00		4.00
Output load factor (ns/pF)	K_{load}		0.27		0.28		0.38		0.57

[1] Parameters have a load dependence (K_{load}), which is used to calculate: $TotalDelay = FixedDelay + (Kload \times Cload)$.

[2] Access time is defined as the slowest possible output transition for the typical and slow corners, and the fastest possible output transition for the fast corner.

Pin Capacitance and Power

❑ Pin capacitance

Pin	Fast@-40C Process 1.98V, -40°C	Fast@0C Process 1.98V, 0°C	Typical Process 1.80V, 25°C	Slow Process 1.62V, 125°C
	Value (pF)	Value (pF)	Value (pF)	Value (pF)
A[j]	0.054	0.054	0.052	0.051
D[i]	0.004	0.004	0.004	0.004
CLK	0.281	0.281	0.270	0.250
CEN	0.015	0.015	0.015	0.014
WEN	0.015	0.015	0.015	0.015
OEN	0.010	0.010	0.010	0.010
Q[i]	0.022	0.022	0.022	0.022

❑ Power

100.00MHz Operation

Condition	Fast@-40C Process 1.98V, -40°C	Fast@0C Process 1.98V, 0°C	Typical Process 1.80V, 25°C	Slow Process 1.62V, 125°C
	Value (mA)	Value (mA)	Value (mA)	Value (mA)
AC Current[1]	7.009	7.100	6.123	5.388
Read AC Current	6.706	6.794	5.860	5.140
Write AC Current	7.311	7.405	6.387	5.637
Peak Current	215.189	204.021	130.094	71.512
Deselected Current[2]	2.337	2.374	1.981	1.701
Standby Current[3]	0.002	0.003	0.002	0.007

[1] Value assumes 50% read and write operations, where all addresses and 50% of input and output pins switch.

[2] Value assumes SRAM is deselected, all addresses switch, and 50% of input pins switch. The logic-switching component of deselected power becomes negligibly small if the input pins are held stable by externally controlling these signals with chip select.

[3] Value is independent of frequency and assumes all inputs and outputs are stable.

Clock Noise and Power/Ground Noise

❑ Clock noise

Signal	Fast@-40C Process 1.98V, -40°C		Fast@0C Process 1.98V, 0°C		Typical Process 1.80V, 25°C		Slow Process 1.62V, 125°C	
	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)
CLK	10.000	0.799	10.000	0.790	10.000	0.818	10.000	0.809

The clock noise limit is the maximum CLK voltage allowable for the indicated pulse width without causing a spurious memory cycle or other memory failure.

❑ Power/Ground noise

Signal	Fast@-40C Process 1.98V, -40°C	Fast@0C Process 1.98V, 0°C	Typical Process 1.80V, 25°C	Slow Process 1.62V, 125°C
	Voltage (V)	Voltage (V)	Voltage (V)	Voltage (V)
Power	0.198	0.198	0.180	0.162
Ground	0.198	0.198	0.180	0.162

The power/ground noise limit is the maximum supply voltage transition allowable without causing a memory failure.

Memory Access Control with FSM

❑ State diagram

- ST0 – Initial state
- ST1 – Write state (write data into SRAM)
- ST2 – Read state (read data from SRAM to evaluate the writing data)
- ST3 – Normal state